

海上回收着陆论文真实性准确性与完整性审查

审查编号: SeaLandingPaper-ResearchIntegrity-Audit-v2

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机器审查结论: PASS_WITH_DECLARED_LIMITATIONS

1. 审查原则

本审查把本项目重新计算的数据、文献对照数据和因公开资料不足而采用的代理输入分开记录。文献曲线只能作为带标签的对照或输入，不能写成本文计算结果；代理几何、数字化缓冲器曲线和数值接触参数不能用于声称复现了原作者的专有模型。任何未通过的数值判据均保留为限制，不以图形平滑或叙述判断改写为通过。

2. 总体结论

当前正文已切换到最新的中等网格水动力、1000 组随机波、实际 6.926 m 四足布局和同一平台四次耦合输出。旧版 1.128 m/m、1.505 m/s、2.435 m/s、2.678 MN 和 0.148 m 等数字均已移除。正文没有把计算未通过项包装为验证成功：仓库四案例认证比较出现 157 个文件级失败，水动力网格判据未通过，22 个随机波工况超出线性适用边界，接触时间步与刚度正则化未通过，526 s 时稳定站立诊断也未通过。

3. 关键结果与证据边界

项目	机器结果	正文解释
中等网格垂荡 RAO	1.113483 m/m	离散频率样本峰值；未称网格收敛
网格判据	FAIL	medium-to-fine 最大选定变化 5.88%，阈值 5%
随机波线性有效	41/63	失效工况不作作业海况或成功率结论
耦合迭代	True	四次固定点的最后一次更新满足 2%
接触时间步	False	峰值力、行程和触地时序不同时收敛
接触载荷资格	False	不得作为真实型号设计载荷
稳定站立	False	动画只能描述当前 20 s 接触过程

4. 正文数值锁定

声明	正文字面值	机器值	来源	存在
yang.stroke_peak	within 4.1%	4.024010180541493	RocketRecoveryCases/Paper_Yang_2026/identified_landing_model/yang-2026-identified-landing-report.json	是
yang_near_optimal	127 of 128	[127, 128]	RocketRecoveryCases/Paper_Yang_2026/identified_landing_model/yang-2026-identifiability-report.json	是

声明	正文字面值	机器值	来源	存在
medium _h eave _{-rao}	1.113 m/m	1.113483	RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/deck-point-rao-medium.json	是
mesh _{-medium-fine}	5.88%	5.879837438854719	RocketRecoveryCases/Barge_120x50/validation/barge-hydrodynamic-convergence.json	是
center _p 95	1.570 m/s	1.570142137428662	RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/wave-sensitivity-revision-600s.json	是
actual _{-feet} p95	1.591 m/s	1.5912185953541447	RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/wave-sensitivity-revision-600s.json	是
valid _{-cases}	41 of 63	[41, 63]	RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/wave-sensitivity-revision-600s.json	是
touchdown _p an _p roduction	0.229 s	0.228999999999985	RocketRecoveryCases/Chrono_LeggedRecovery/chrono-same-platform-multibody-report.json	是
production _p eak _{-force}	5.006 MN	5.006461513957208	RocketRecoveryCases/Chrono_LeggedRecovery/chrono-same-platform-multibody-report.json	是
platform _{-energy-residual}	-6.43e-6	-6.432821992171707e-06	RocketRecoveryCases/Chrono_LeggedRecovery/chrono-same-platform-multibody-report.json	是
cert _{-file-failures}	157 file-level failures	"Cylinder": 65, "DeepCwind": 13, "HywindSpar": 7, "Moonpool": 72	analysis/diagnostics/hams-cert-root-cause.json	是

5. 文件溯源

文件	类别	用途	存在	SHA-256 前缀
新 论 文 参 考 写 作 论 文 /Analysis+of+Sea-Based+Landing+Dynamics+of+Reusable+Landing+Vehicle+Considering+Mechanism+Flexibility.pdf	reference _{source}	Yang et al. paper	是	e7949d55dea5a5d9
新 论 文 参 考 写 作 论 文 /Analysis+of+Sea-Based+Landing+Dynamics+of+Reusable+Landing+Vehicle+Considering+Mechanism+Flexibility.md	reference _{source}	Yang et al. extracted text	是	3939e6f442f1738d
海上平台火箭回收文献/065002_1_2.0002061.pdf	reference _{source}	Nargolkar and Vijayan paper	是	f9ba2eb6197a8ad7
海上平台火箭回收文献/P3_Thies_2022.pdf	reference _{source}	Thies landing-leg paper	是	98d22ac4d4995872
海 上 平 台 火 箭 回 收 文 献 /P2_WangZhi_2023_ShipSciTech.pdf	reference _{source}	Wang plume/platform paper	是	afd1269bd8d00a75
RocketRecoveryCases/Paper_Yang_2026/reference/digitized/yang-2026-fig09-11-digitization.json	reference _{target}	Digitized published test traces	是	309fb9da09933f8c
RocketRecoveryCases/Chrono_LeggedRecovery/validation/thies_absorber_curves/thies-buffer-curves.json	reference _{target}	Digitized absorber curves and axis calibration	是	e4971045da507e89
RocketRecoveryCases/Barge_120x50/platform_config.json	present _{input}	Parameterized barge input	是	50f90e7fbd6f1f1b
RocketRecoveryCases/Barge_120x50/validation/barge-self-check.json	present _{calculation}	Analytic mesh and hydrostatic self-check	是	06359d8905263bfc
RocketRecoveryCases/Barge_120x50/validation/barge-hydrodynamic-convergence.json	present _{calculation}	Hydrodynamic refinement, A-infinity and IRF checks	是	69304f742c7bf2ca
RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/deck-point-rao-medium.json	present _{calculation}	Medium-grid local deck response operators	是	61091ab1919a2593
RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/wave-sensitivity-revision-600s.json	present _{calculation}	1000-realization random-wave calculation	是	5519de6bbc861e8c

文件	类别	用途	存在	SHA-256 前缀
RocketRecoveryCases/Barge_120x50/Output/RocketRecovery/wave-duration-sensitivity.json	present_calculation	600/1200/1800 s duration sensitivity	是	4392ba5f0490553f
RocketRecoveryCases/Paper_Yang_2026/identified_landing_model/yang-2026-identified-landing-report.json	present_calculation	Reduced-model comparison	是	da1f143fe0d4051e
RocketRecoveryCases/Paper_Yang_2026/identified_landing_model/yang-2026-identifiability-report.json	present_calculation	128-start identifiability analysis	是	98f77e5b97f6558a
RocketRecoveryCases/Chrono_LeggedRecovery/chrono-same-platform-multibody-report.json	present_calculation_with_labelled_proxies	Same-platform four-pass multibody report	是	868e066a08d712f2
RocketRecoveryCases/Chrono_LeggedRecovery/Output/RocketRecovery/chrono-same-platform-multibody-response.json	present_calculation_with_labelled_proxies	Full same-platform response history	是	eb6546558296c4ee
RocketRecoveryCases/Chrono_LeggedRecovery/chrono-revision-study-report.json	present_code_verification	Independent smooth-law energy/code companion	是	0954aa98bbbee771
analysis/diagnostics/hams-cert-root-cause.json	failed_external_regression	Official four-case comparator diagnostic; no validation credit	是	03d9bb625c5540f2
paper/yang_2026_extension/manuscript.md	present_manuscript	Submission manuscript source	是	041a9e65806ae910
analysis/diagnostics/reviewer-spectrum-audit.json	present_calculation	Archived-grid full-spectrum normalization sensitivity	是	0613a1895ef2e83b
analysis/diagnostics/reviewer-matrix-audit.json	present_calculation	Raw hydrodynamic reciprocity and damping eigenvalues before symmetrization	是	0353c516aef8311c

6. 图件审查

全部图件由同一脚本基于机器输出生成，采用 SciencePlots science/no-latex 风格、Times New Roman 字体，并同时输出 600 dpi PNG 和可编辑 SVG。文献曲线使用灰黑虚线，本文计算使用彩色实线。图形不经办公软件手工移动数据点。

图件	用途	像素	TNR	SHA-256 前缀
fig01-coupled-framework	Implemented theoretical data flow	4299 x 2019	是	0bd63483100ef73a
fig02-barge-geometry-mesh	Present geometry and two deck-point layouts	4029 x 2257	是	6b271ae6b1f9f857
fig03-yang-identified-landing-comparison	Digitized reference versus present reduced-model calculation	4296 x 3399	是	0a5197c871668851
fig04-platform-hydrodynamic-response	Present hydrodynamic and refinement calculations	4299 x 2799	是	db83a3110b4e5a39
fig05-random-wave-deck-statistics	Present 1000-realization statistics and validity flags	4299 x 2799	是	66f2168bdcafa5c
fig06-partitioned-contact-feedback	Present same-platform multibody response and numerical sensitivity	4299 x 4719	是	b4d06de4475e7e77
figS01-yang-identifiability	Present multi-start identifiability result	4303 x 1869	是	495d07339611b49b
figS02-yang-deck-filter-reproduction	Supplementary transfer-function reconstruction	4299 x 2769	是	a68a77a375f853a7

7. 未解决问题及结论约束

严重度	项目	证据	结论约束
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严重度	项目	证据	结论约束
high	Repository certification regression	The registered four-case comparator reports 157 failed output files (65/13/7/72 by case).	No solver-certification or hydrodynamic-benchmark pass is claimed; cause remains unresolved.
high	Hydrodynamic mesh refinement	Medium-to-fine maximum selected change is 5.88%, above the 5% criterion.	Medium-grid RAOs are reported as sampled case-study results, not mesh-converged predictions.
high	Linear free-surface validity	Only 41/63 random-wave cases pass tilt and deck-edge criteria; worst P95 tilt is 11.99 deg.	Failed conditions are screens for nonlinear analysis, not operational predictions.
high	Contact numerical convergence	Time-step convergence and contact-stiffness regularization both fail for local force/stroke/touchdown metrics.	No contact peak is qualified as a design load.
high	Stable standing	Final vertical speed and angular rate exceed the diagnostic thresholds.	The simulated interval demonstrates contact dynamics, not a completed stable landing.
medium	Reduced-model identifiability	127/128 starts are near-optimal and parameters are strongly correlated.	Yang curves support response comparison, not unique physical parameter recovery.
medium	Horizontal platform closure	Fx, Fy and Mz are calculated but not injected because horizontal restoring is absent.	Feedback is restricted to heave, roll and pitch.
medium	Literature-derived full-scale inputs	Vehicle geometry and absorber curves are proxies digitized from public literature; azimuths and component inertia allocation remain declared assumptions.	The full-scale calculation is a proof of concept, not a reproduction of proprietary vehicle data.

8. 可重复执行命令

```
python .\analysis\rocket_recovery\barge_hydrodynamic_convergence.py --execute
python .\analysis\rocket_recovery\deck_point_rao.py --case .\RocketRecoveryCases\Barge_120x50
python .\analysis\rocket_recovery\barge_wave_sensitivity.py report
python .\analysis\rocket_recovery\barge_wave_duration_sensitivity.py report
python .\analysis\rocket_recovery\yang_2026_identifiability.py report
python .\analysis\rocket_recovery\chrono_same_platform_multibody.py report
python .\analysis\rocket_recovery\yang_2026_figures.py
python .\analysis\rocket_recovery\paper_integrity_audit.py report
```

9. 审查结论

PASS_WITH_DECLARED_LIMITATIONS 表示数字、来源、代码、图件和正文限定语之间已建立可追溯关系，不表示所有数值模型均通过验证。现阶段能够支持的是理论接口、实现可重复性、局部甲板运动机制和同一平台反馈演示；不能支持真实型号支腿载荷、着陆成功率、允许海况或全六自由度动力定位闭环。